

MAX ZHU

ATLANTA, GA | MAX.ZHU@GATECH.EDU | 404-940-3568 | MAXZHU143.GITHUB.IO

EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S in Computer Engineering

Expected Graduation December 2026

- Threads: Computing Hardware and Emerging Architecture and Systems and Architecture

GPA: 3.9/4.0

TECHNICAL SKILLS

Software: C/C++, Python, Perl, TCL, RISCv, Javascript, Java, MatLab, HTML, MySQL, Git

Hardware: FPGA, Verilog, UVM, Oscilloscopes, Digital Circuits, VHDL, KiCad, Embedded System, Arduino, CAD

EXPERIENCE

CRNCH Rogues Gallery

Atlanta, GA

Reconfigurable Researcher

January 2025 – Present

- Using **Verilog** to create the XChaCha20 encryption algorithm and **Docker** to deploy the package on a **PYNQ** board
- Offloaded cryptographic operations from the CPU to a **PYNQ FPGA**, seeing up to 10 times higher efficiency

MARGA Lab

Atlanta, GA

Wireless Digital Twin System Researcher

August 2024 – Present

- Analyzed and annotated 50+ research papers on **5G, Wi-Fi 6, ray tracing, wireless signals, and UAVs**
- Created scripts in **Python** using **Sionna** to simulate and optimize transmitter-receiver placement by over 60%
- Manipulated environments in Blender to reduce workload by 50% while maintaining over 95% accuracy.

Metro Healthcare

Vancouver, BC

Frontend Developer

May 2024 - August 2024

- Worked in an **Agile** project environment and continuously progressed by adapting to backend additions
- Developed an interactive UI for 15+ pages with React.js to showcase clinical services and purchasable products

EXTRACURRICULARS

Silicon Jackets | Digital Designer and Verification Engineer

August 2024 – Present

- Designed and implemented an **arithmetic logic unit** for a CPU from scratch with **Verilog**
- Developed and synthesized over 100+ tests with **Python** to verify correct hardware functionality

HIVE | Master Peer Instructor

December 2023 – Present

- Led workshops and guided users laser cutting, 3d printing, wood working, and **PCB fabricating**
- Fixed and maintained sewing, PCB creating, and wood working machines,

PROJECTS

RISCv Processor | System Verilog, GTKWave

- Created a 5-stage **RISCv** pipelined processor with **System Verilog**, complete with forwarding and stalling controllers
- Debugged waveforms with **GTKWave** and handled structural, data, and control hazards

SCOMP Processor and External Memory Device | VHDL, DE10 Standard, SCOMP

- Created and documented a Simple Computer(SCOMP) processor and custom memory device peripheral with **VHDL**
- Demonstrated correct functionality of all components with a **DE-10** FPGA board

Spotify Car Thing and Wrapped App | ESP8266, C++, Django, Python, HTML5, CSS

- Created a DIY Spotify Car Thing replica with an **ESP8266**, **Spotify Developer API**, **C++**, and physical components
- Connected to the **Spotify Developer API** and used tokens for **OAuth** to grab live-playback and user song data
- Created custom profile analysis with **AI** and slide animations with **CSS** and **Javascript** for a personalized presentation
- Deployed the website with **Heroku** handling pipeline conflicts while securely handling user data

Cloud Server | Linux CLI, Docker

- Configured and deployed a cloud server with **Ubuntu Server** to replace multiple personal subscription services
- Exclusively utilized **Linux CLI** (command-line interface) for system administration, server organization, and **automation**